

Expressive Italian TTS for News Narration

Fine-Tuning VITS, collecting data, inference tweaks & more

Nikolai Gorbachev
DHDK UniBo

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Introduction

Introduction: Why Italian TTS?

- TTS (Text-to-Speech) is becoming a key part of human–computer interaction
- English has many advanced systems, but Italian TTS is still scarce
- Expressive, natural-sounding voices are needed for real-world use cases
- Personally: I'd love to use Italian TTS to improve my language skills

Introduction: Why TTS for News Narration?

- Reading newspapers is fading, but listening is growing
- Commutes & daily routines create natural time for audio
- Audiobooks are too complex to model → news narration is a realistic first step
- Goal: build a moderately expressive, human-like Italian TTS that can be trained even in Colab

Exploring Existing Models

Exploring Existing Models

Model	✓ Pros	⚠ Cons	Open-Source?
Parler-TTS	Multilingual, style prompting	Very slow, no voice cloning	Yes / research focus
Bark	Expressive, high quality	Slow, limited control	Yes / commercial restrictions
xTTS2	Best prosody, very natural	Sometimes too expressive, license issues	Partially / commercial restrictions
VITS Baseline	Understandable, fast, basic prosody	Robotic, flat intonation, poor stress control	✓ Fully open-source



Exploring Existing Models

Models tested:

- Parler-TTS
- Bark
- XTTS2

Key Observation:

- All three give good results, but lack fine-grained control over voice, intonation, and stability

Takeaway:

→ Motivates fine-tuning our own model (VITS) on curated Italian news data

Data Collection and Enhancement

Data Collection

Source: *Il Resto del Carlino* TG playlists

Audio Download: WAV using [yt-dlp](#)

Crop & Clean: remove intros/outros, separate vocals with Spleeter

Segmentation: ~10s chunks via silence detection, min 1s

Transcription: Italian transcripts with OpenAI Whisper

Output: [wav](#)|[text](#) metadata ready for VITS fine-tuning



Data Enhancement

1. Punctuation Restoration

- Many transcripts lacked proper punctuation → robotic prosody
- Used **multilingual punctuation restoration model**
- Result: more **natural pauses** and expressive speech

2. Audio Padding

- Observed glitches at the start of clips (~60k steps)
- Added **100ms silent padding** at the beginning of each audio file
- Result: smoother intonation, fewer initial artifacts



VITS: Architecture overview

VITS Architecture: key points

Latent Audio Representation (z)

- Probabilistic vector per time step
- Captures pitch, timbre, and prosody variations

Variational Inference (VI)

- Posterior $q(z|x, y)$ aligns with prior $p(z|x)$
- Generates realistic audio from text

Normalizing Flows

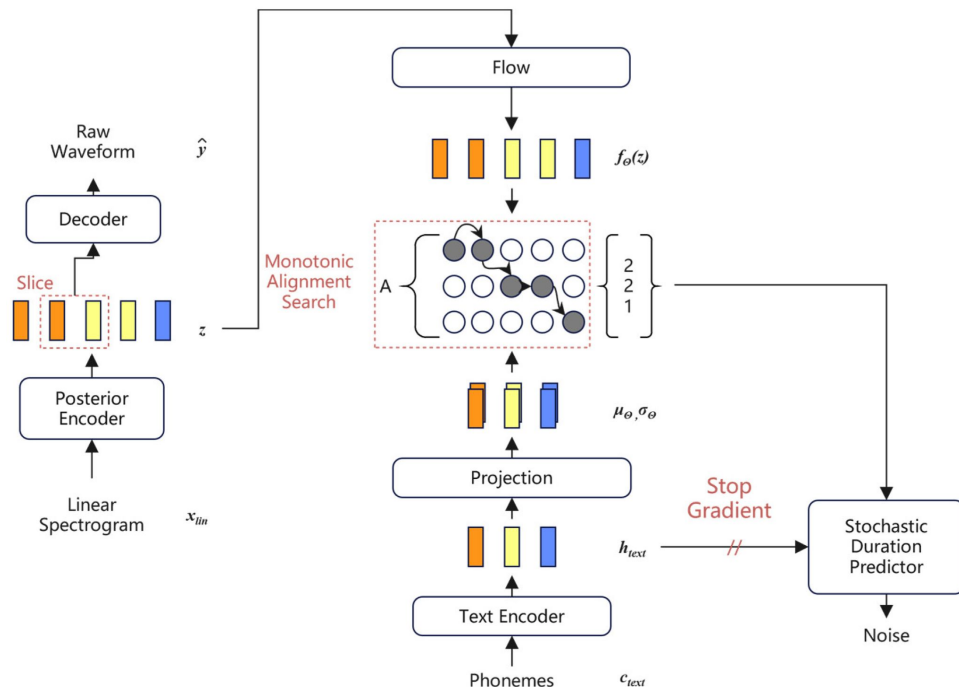
- Transforms simple prior $\rightarrow$ complex latent distribution
- Enables expressive audio sampling

HiFi-GAN Decoder

- Renders waveform from latent z , per time step
- Produces smooth, natural audio

Monotonic Alignment Search

- Aligns text tokens with latent audio segments
- Avoids explicit duration prediction



VITS Architecture: advantages

Conventional TTS

Predicts mel spectrogram → needs separate vocoder

Deterministic output

Cascaded errors: mel → vocoder

Often slower to train

VITS

End-to-end latent-to-waveform

Probabilistic latent: natural variation

Single model: fewer error cascades

Efficient, high-quality synthesis

VITS: Information flow

1. **Text input** → fed into **Text Encoder** → produces text embeddings
2. **Latent prior $p(z|\text{text})$** is computed from text embeddings
3. **Audio Encoder** processes the reference audio → produces posterior $q(z|\text{text}, \text{audio})$
4. **Normalizing Flow** transforms prior → posterior for flexible latent representation
5. **Sample latent z** from posterior/prior
6. **HiFi-GAN Decoder** converts latent z → waveform
7. **Waveform output**: final generated audio

VITS: Finetuning and Inference

VITS Finetuning

VITS Fine-Tuning

Load Pretrained Model

- Italian VITS ([tts_models/it/mai_male/vits](#))
- Pretrained checkpoint & config
- Coqui-TTS / IDIAP implementations

Dataset Preparation

- WAV files + metadata ([wav|text|text](#))
- Audio processed, segmented, padded, punctuated

Observations Before Fine-Tuning

- Works out-of-the-box for Italian
- **Strengths:** understandable pronunciation
- **Weaknesses:** robotic, flat prosody, wrong pauses, no accented letters



VITS Training Timeline & Observations

- **Before ~20k steps:** first understandable Italian speech, missing punctuation → restored punctuation
- **~60k steps:** padding issues at audio start, stress errors → apostrophe used as stress mark
- **~80k steps:** plateau → stopped to avoid overfitting on 1.5h dataset



20k



30k



50k



65k



75k



80k

VITS Inference Tweaks

Text Prompt Tweaks

- Accents, apostrophes, double letters
- Italian spelling and lowercase letters



Effect:

- More natural prosody
- Better pronunciation
- Minor exaggerated stresses

Audio Padding Tweaks:

- Small leading pause at beginning of audio
- Improves rhythm and smoothness

VITS Inference Tweaks

Reference Voice Sample:

- Mimics intonation and style of sample
- Combines with text and padding tweaks



Effect:

- More controlled style
- Less accurate stresses
- Works together with punctuation & apostrophes for final polish

VITS vs XTTS2 (WER Evaluation)

VITS vs XTTS2 – WER Evaluation

VITS model: Finetuned Italian VITS (~1.5h data, 80k steps)

Comparison model: xTTS2 multilingual TTS

Evaluation metric: WER (Word Error Rate) on short Italian audio chunks

Transcription tool: OpenAI Whisper (**small**) with number normalization

$$W = (S + D + I) / N$$

S = number of substitutions (wrong words)

D = number of deletions (words that were missed)

I = number of insertions (extra words that were not in the reference)

N = total number of words in the reference transcript

WER Results

File	VITS	xTTS2
chunk_00900.wav	0.394	0.121
chunk_00700.wav	0.722	0.111
chunk_00600.wav	0.667	0.111
chunk_00500.wav	0.455	0.182
chunk_00300.wav	0.273	0.364

Conclusion

Final Thoughts - Results Overview

Subjective Evaluation

- Speech quality: **good**, understandable, reasonably natural
(unlike most open-source Italian models)

Open-Source & Fast

- Finetuned Italian VITS is **open-source**
- **Much faster inference** than other open-source models

Performance

- **Much better than baseline VITS**
- Still behind xTTS2 in WER
 - xTTS2: licensing restrictions, slightly worse intonation control

Final Thoughts - Results Overview

Evaluation

- Explore **quantitative metrics** for stylistic naturalness and speech quality

Data & Style

- Increase **dataset size** carefully to avoid overfitting
- Include **stylistically versatile data** to improve reference audio prompting

Model & Architecture

- Investigate **genre/style tagging** (tokens or characters)
- Consider **architecture tweaks** for better prosody or style control

Takeaway

Model is usable, open-source, and a strong baseline for further experimentation



Thank you for your attention!